STEEL ERECTION

MES-ST-SPC-0002

PROPRIETARY & CONFIDENTIAL

This document and the information contained herein is Ma'aden proprietary know how. This document or any part thereof should not be reproduced, stored in a retrieval system, or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

REV NO.	REV DATE	DOC NO.	REVIEW CYCLE	REVIEW DATE	PREPARED BY	REVIEWED BY	VERIFIED BY	APPROVED BY (EMPLOYER)
C02	02-JUL-24	MES-ST-SPC-0002	5 Years	02-JUL-29	SS	TK 2024.07.02 09:03:39 +02'00'	Digitally signed by MID61704 Date: 2024.07.05 07:36:32 +05'30' eWc 2024.07.05 08:15:35 -05'00'	AAG

REVISION HISTORY

REV NO.	REV DATE	PURPOSE	PREPARED BY	REVIEWED BY	VERIFIED BY	APPROVED BY (EMPLOYER)
C01	28-SEP-11	Issued for Use	K H Chong	D Kirkwood	XX	P Burrows
C02	02-JUL-24	Issued for Use	SS	TK	eWc	AAG

REV NO.	REV DATE	REVISION HISTORY
C01	28-SEP-11	Issued for Use
C02	02-JUL-24	Revised as indicated below: 3.2.2.3 -ASSE changed to ASSP- American Society of Safety professionals 3.2.4- MD-101-SMEM-EG-ST-SPC-0002 last number corrected to 0001. 5.2.1- Last para is deleted 5.2.2- ASSE is changed to ASSP 5.3.1- Last para qualified with "However all such pre-assembly plans shall be reported to the employer in advance for review. Erection shall be commenced only after receipt of acceptance from employer. 5.3.4- Threaded rod qualified with " Post-installed threaded rods" 5.3.5- First para qualified with "or when indicated on design drawings. Spec No corrected to "0001" from "0002." 5.3.6- Sixth para "Frame cutting "corrected to "Flame cutting." 6.2.1- Additionally qualified with "It shall be approved by the employer for such cases"

Table of Contents

1. Introduction 4

1.1 Scope 4

1.2 Purpose 4

1.3 Conflicts and Deviations 4

2. Key Terms and Definitions 4

3. Referenced Documents 4

3.1 Statutory Regulations 5

3.2 Industry Codes and Standards 5

3.2.1 Local Codes and Standards (for Saudi Arabia) 5

3.2.2 International Codes and Standards 5

3.2.3 Alternative Design Codes 6

3.3 Ma'aden Standards 6

3.4 Site Data 7

4. Handling, Shipping, Delivery and Storage 7

4.1 Handling and Shipping 7

4.2 Delivery 7

4.3 Storage and Handling 7

5. Steelwork Erection 8

5.1 General 8

5.2 Erection Safety 8

5.2.1 General 8

5.2.2 Static Line Anchorage 9

5.2.3 Verification of Supports and Anchor Bolts 9

5.2.4 Pre-Erection Checks 9

5.3 Erection Procedure 10

5.3.1 General 10

5.3.2 Propping and Temporary Works 11

5.3.3 Packers and Shims 11

5.3.4 Bolt Connections 11

5.3.5 Site Welding 12

5.3.6 Tolerances 12
5.3.7 Grouting of Bases 13
5.3.8 Steel Joist 13
5.3.9 Floor Plate, Grating and Handrails 14
5.3.10 Coating Repairs 14
6. Field Modifications 14
6.1 General 14
6.2 Fabrication 14
6.2.1 Cutting 14
6.2.2 Holing 14
6.2.3 Repositioning 15
6.2.4 Finishing 15
6.2.5 Temporary Bracing Supports 15
6.2.6 Removal of Gussets 15
6.2.7 Plugging of Bolt Holes 15
7. Inspection and Testing 16
7.1 General 16
7.2 Inspection Requirements 16
7.2.1 Materials 16
7.2.2 Survey 16
7.2.3 Surface Treatment 16
7.2.4 Field Welding 16
7.2.5 Bolting 17

1. INTRODUCTION

1.1 Scope

This document provides the specification for erection of structural steelwork for Ma'aden Projects.

This document is intended to be used in conjunction with the applicable documents listed herein.

1.2 Purpose

The purpose of this specification is to set the minimum standards and requirements for the receipt, handling, erection, assembly, and field inspection of structural and miscellaneous steel.

1.3 Conflicts and Deviations

Any deviation from the requirements stipulated in this document shall go through a waiver approval process as per Ma'aden Capital Investment System (MCIS) guidelines and shall be addressed to Ma'aden's Project Development & Engineering (PDE) team for their review and approval.

2. KEY TERMS AND DEFINITIONS

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates
Contractor	Party responsible for supplying materials, plant, labor and supervisory services
Manufacturer	Party who produces and warrants performance of products,materials, and/or items provided in accordance with contract documents
Supplier	Party responsible for supplying concrete materials in accordance with contract documents

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable. In general, the Contractor shall be responsible for full compliance with applicable Ma'aden Health, Safety and Environmental Standards (MSHEMs). Any deviation from the MSHEMs shall require written approval from the Employer.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia or local requirements of Project Site. The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

The work shall conform to the following applicable codes and standards.

3.2.1 *Local Codes and Standards (for Saudi Arabia)*

3.2.1.1 Saudi Building Code

All Ma'aden facilities shall comply with latest version of all Saudi Arabia applicable local codes and standards including but not limited to:

SBC 306	Structural – Steel Structures.

For project locations other than Saudi Arabia, the Contractor shall follow the respective applicable local regulations, codes and standards

3.2.2 *International Codes and Standards*

3.2.2.1 American Institute of Steel Construction (AISC)

AISC 360	Specification for Structural Steel Buildings
AISC 326	Detailing for Steel Construction
AISC 303	Code of Standard Practice for Steel Buildings and Bridges
	AISC Manual of Steel Construction, Allowable Stress Design (ASD)
	AISC Specification for Structural Joints Using ASTM A325 or A490 bolts

3.2.2.2. American National Standard Institute (ANSI)

ANSI A10.13 Steel Erection – Safety Requirements

3.2.2.3. American Society of Safety Professionals

ASSP FALL PROTECTION Introduction of Fall Protection

3.2.2.4. Process Industry Practices (PIP)

STS05130 Erection of Structural and Miscellaneous Steel Specification

3.2.2.5. American Welding Society (AWS)

AWS D1.1 Structural Welding Code – Steel

3.2.2.6. Steel Deck Institute (SDI)

ANSI/SDI SD Standard for Steel Deck

ANSI/SDI C Standard for Composite Steel Floor Deck

SDI Manual of Construction with Steel Deck- No.MOC3

3.2.2.7. Steel Joist Institute (SJI)

SJI Standard Specifications and Load Tables

3.2.2.8. U.S Department of Labor, Occupational Safety And Health Administration (OSHA)

OSHA 29 CFR Part 1926 Safety and Health Regulations for Construction

3.2.3 *Alternative Design Codes*

Code of Practice – Prevention of Falls at Workplaces – 2004, published by the Commission for Occupational Safety and Health – Western Australia

3.3 Ma'aden Standards

MD-101-SMEM-EG-ST-SPC-0001 Steelwork Fabrication

MD-101-SMEM-EG-ST-SPC-0003 Grout

MD-101-SMPM-PM-QA-PEI-0001 Contractor Quality Requirements

3.4 Site Data

All materials, equipment and work methods shall be suitable for the Site conditions as defined in the applicable Site Data Specification.

4. HANDLING, SHIPPING, DELIVERY AND STORAGE

4.1 Handling and Shipping

Delivery of steel shall be made in the order needed for erection. The delivery sequence for the fabricated steel, unless otherwise noted in the Contract Documents, shall be in accordance with the schedule of the particular steelwork erection.

All bolts, washers and nuts shall be packaged and delivered in rigid, weatherproof containers.

The contractor shall ensure that all steel and its coatings are protected from any damage caused by handling, storage or shipping before receipt.

Contractor shall ensure that adequate protection is provided for threads on sag rods and any other threaded components to prevent damage during shipping and handling.

4.2 Delivery

All materials and documentation will be inspected immediately upon receipt to determine that all items included in the Bill of Materials have been supplied, to assure that all documentation has been received and to check for any damage. Employer reserves the right to reject all damaged or below-quality material or documentation.

4.3 Storage and Handling

All necessary supervision, chains, timber dunnage, tie-down equipment, flags, permits and the like shall be provided to safely and carefully complete transportation of structural steel.

Structural steel for erection shall be stacked clear of the ground and be self-draining.

Wooden supports in contact with finished items during transport and storage shall be covered with a soft material such as rubber to minimize damage to the coating. Projections from steelwork such as lugs and splice plates which may suffer damage during transport are required to be reinforced with a timber packing piece.

Hessian fabric or carpet shall be used between the painted items and holding down chains to prevent damage to the paint coating, during transportation.

All steelwork that has been finish coated shall be handled with care so as to preserve the coating in the best practical condition. Slings used for the handling of the finished steelwork shall be covered with soft material.

All small loose items including bolts, nuts and washers shall be supplied in grit-free containers and shall be kept in water-tight storage. Each box shall be identified with a securely fastened metal tag bearing its identification number and the approximate reference of its intended use.

Any damage sustained during handling shall be reported to the Employer immediately once it is discovered, and corrective actions shall be undertaken with the approval of the Employer or shall be replaced with NO liability or extra cost to the Employer.

5. STEELWORK ERECTION

5.1 General

Erection shall comply with this specification, Drawings, and the relevant codes, in particular AISC 360. All work shall be carried out by qualified riggers and tradesmen using the best and safe working practices, in accordance with OSHA 29 CFR PART 1926, Safety and Health Regulations for Construction.

5.2 Erection Safety

5.2.1 *General*

A Steel Erection Safety Plan shall be provided, detailing methods and procedures to be employed and the responsible personnel, to ensure that the requirements of this Specification and the applicable codes, statutory regulations and standards are met.

As part of the Safety Plan, an Inspection Plan shall be provided for all major elements of the work detailing inspection procedures, responsible personnel and safety programs.

The Inspection Plan shall identify priorities at the critical milestones in the work process. Work shall not proceed until an acceptable program has been approved by the Employer.

Safety inspections shall be carried out in accordance with the approved Safety Plan. It is the Contractor's responsibility to provide safe access and egress to work areas and a safe system of work so that, as far as practicable, employees are not exposed to hazards.

All rigging equipment used during erection must comply with and be certified to meet local or national safety requirements. The Contractor shall be responsible for ensuring that all lifting or

support equipment is supplied, positioned, and used in such a way that the loads taken are at all times within the safe operating limits of the equipment being used and of the ground or supports on which the equipment rests.

5.2.2 *Static Line Anchorage*

Static line anchorage points for fall arresting devices shall be provided on all structural steelwork, in accordance with ASSP FALL PROTECTION. However, the Contractor shall design and install a fall arrest system that is customized to account for the requirements of each individual location, for all variables such as number of personnel, length, support spacing and height of installation. In case of the Contractor designing a fall arresting system, it has to be duly submitted for approval by the Employer before commencement of the work.

5.2.3 *Verification of Supports and Anchor Bolts*

Prior to commencing erection, the following verifications shall be carried out:

- The actual location and elevation of all points of support of the steel structure shall be checked against structural steel drawings by the Contractor Surveyors/Engineers and then inspected by the Employer's Surveyors/Engineers in the field.
- It shall be confirmed that a minimum clearance, as nominated on the drawings, is available for grouting between the concrete footing or pedestal and base plate or any other section of steelwork requiring grouting, and
- The size, location, elevation, and projection of anchor bolts shall be checked against structural steel drawings. Anchor bolts shall be checked to ensure that they are vertical and that threads are clean and undamaged.

Any discrepancies or damage to the concrete supports shall be reported to the Employer before any steel erection commences.

Steelwork may be placed on concrete supports only after the 7-day compressive strength results are found to be satisfactory.

5.2.4 *Pre-Erection Checks*

All structural members shall be visually checked for compliance with the fabrication tolerance requirements of Steel Fabrication Specification, to ensure that no damage or distortion to steelwork, coatings or fireproofing has occurred during transit to site. The Employer reserves the right to check the members for conformity thoroughly and reject material if found non-conforming.

All such distortion or damage shall be reported to the Employer before accepting the steelwork for erection.

Surfaces in contact with other steelwork or concrete shall be cleaned to remove all loose mill scale, rust, oil, grease, dirt, weld splatter, slag and any other foreign matter.

The surface of the foundation under any section to be grouted, shall be chipped, and brushed by the Contractor to remove the weak upper layer of the concrete, damaged concrete and any oil-soaked concrete, before steel is erected on the foundation.

5.3 Erection Procedure

5.3.1 *General*

Erection shall be in accordance with the drawings and other Contract Documents, AISC Manual of Steel Construction (ASD and LRFD), the AISC Code of Standard Practice for Steel Buildings and Bridges, OSHA 29 CFR Part 1910 and Part 1926, and SBC 306.

Any circumstances discovered by Contractor that affect progression, performance, or completion of Contractor's work activities such as discrepancies between the erection/shop drawings and the delivered steel members, incorrectly fabricated steel members or incomplete or unacceptable work of other contractors affecting Contractor's work shall be immediately reported in writing.

Erection of steel joists shall conform to the requirements of the Steel Joist Institute (SJI), the joist manufacturer, this code/specification, and OSHA 29 CFR 1926 Subpart R.

Care shall be taken to avoid damage to steel coatings.

Before final bolting or welding, erected structures are to be checked dimensionally for line, level, alignment and plumb. Erected structures are to be re-checked for line, level, alignment and plumb after final bolting or welding.

The Contractor shall submit a request for inspection to the Employer after the completion of each stage of erection activities and initial check. The Employer may check for line, level, alignment and plumb of the erected structures before giving clearance for the next stage of erection.

When major steelwork members are lifted, they are to be slung or supported in such a manner as to prevent undue distortion or flexing of the member during lifting operations.

It is permissible to partially assemble structures before lifting into position provided all necessary precautions are taken to prevent distortion, flexing or undue stresses in the structure during lifting

operations. However, all such pre-assembly plans shall be reported to the employer in advance for review. Erection shall be commenced only after receipt of acceptance from the Employer.

5.3.2 *Propping and Temporary Works*

The contractor shall be responsible for the adequacy and installation of any temporary bracing or guy cables required to counter-act loadings imposed during erection. This responsibility shall also extend to temporary bracing required to ensure safe and stable conditions of partially completed structural assemblies. The structure shall be plumbed, leveled, and braced before any final bolted or welded connections are made.

Approval shall be obtained from the Employer before any welding of temporary works to the structural steelwork and before drilling or other modifications to the structural steelwork to accommodate temporary works.

Welding shall be carried out in accordance with approved welding procedure. Removal of any attachment shall be carried out by cutting 3 mm above the surface and grinding level with the surface. Surface of the affected area shall be examined for cracks by dye penetrant or magnetic particle examination.

No major temporary braces, supports or equipment shall be mounted on the steelwork structure without the approval of the Employer.

5.3.3 *Packers and Shims*

Plate packers and shims shall be provided to "shim" base plates to the correct level for grouting.

The packing shall be of an adequate size and in sufficient number to suit the size of the base plate. Packers shall be made from galvanized mild steel plate (or other material as approved by the Employer), and shall be clean, flat and free from burrs or other imperfections. Number of packers per pack shall be no more than three.

Packers and shims shall be positioned under the base plate so as to be fully encased in the grout envelope. Grout cover to the plates at the edges shall be at least 50 mm.

Packers and shims shall not be used in joints made using high strength structural bolts unless they are shown on the fabrication drawings and are supplied with the steelwork.

5.3.4 *Bolt Connections*

Installation and method of tensioning of Snug Tight Type, Pre-Tensioned Type and Slip-Critical Type bolts shall be in accordance with AISC, using either:

- Part Turn Method of tensioning
- Tensioning by use of a Direct Tension Indicating Device.

The proposed method of tensioning bolts shall be checked and verified on site and the proposed procedure shall be submitted to the Employer for approval, prior to commencement of work.

Adjustable spanners and pipe wrenches shall not be used for the installation of structural bolts.

The contractor shall color code, die punch or otherwise mark the ends of the torqued bolts indicating that the bolts have been properly tensioned and are ready for inspections.

Post installed threaded rods anchored into concrete shall be anchored with Hilti HIT H 150 Adhesive Anchors or approved similar. Preparation installation and curing shall be accordance with the Manufacturer's instructions. All rods, nuts and washers shall be galvanized.

5.3.5 *Site Welding*

Site welding may be used when required due to discrepancy, inaccuracy, or for the purposes of field fabrication or when indicated on design drawings

All welding on site shall be carried out in accordance with the requirements stated in the Project Specification, MD-101-SMEM-EG-ST-SPC-0001, Steel Fabrication Specification. All relevant details and proposed procedures shall be approved by the Engineer prior to the commencement of any site welding.

Adequate access and mobile welding facilities shall be provided for any site welding operations.

Welding shall conform to AWS D1.1. All welds shall be continuous unless an alternate procedure is approved.

Minimum fillet weld size shall be 5 mm for structural welds. Seal welds may be 3 mm minimum fillet weld.

Welds to connection plates embedded in concrete shall be deposited in a sequence that minimizes distortion of the embedment to ± 3 mm of flat or true.

5.3.6 *Tolerances*

Erection tolerances given in SBC 306 shall not be exceeded.

Centre-to-centre distances of bolt holes (including diagonal and radial distances) for equipment setting, shall not exceed ± 2 mm of those nominated.

Supports for crane runway beams shall be within ± 3 mm of nominated elevation. Rails for gantry cranes shall not deviate more than ± 3 mm from the nominated line, with no discontinuities at rail joints.

Fit-up bolts shall be used to bring improperly fabricated members and parts into place. Drift pins shall not be driven with such a force as to damage adjacent metal areas.

When approved Contractor may enlarge standard holes by 1 mm when necessary to make connections resulting from minor misfit. Holes in connections that misfit by more than 1 mm shall be corrected as directed.

Enlargement of holes shall be made by reaming or drilling only. Flame cutting, burning, gouging, chipping or drift punching shall not be permitted.

No packing, shimming, filling or wedging shall be permitted to correct faulty work unless approved.

One filler plate up to 3 mm thickness may be used as required in spaces between members to be bolted. Approval shall be required for filler plate requirements greater than 3 mm.

5.3.7 *Grouting of Bases*

All procurement, supply and installation grouting materials shall be in accordance with Project Specification, MD-101-SMEM-EG-ST-SPC-0003, Grout.

After plumbing and alignment of steelwork and final bolting or welding, and with the prior written approval of the Employer the structure shall be grouted in accordance with the above referenced Specification.

Unless otherwise approved by the Employer, all grouting shall be complete and cured prior to the application of loads which may cause damage due to ungrouted base plates or supports, e.g. concrete floors, hydrostatic testing of tanks or pipelines.

5.3.8 *Steel Joist*

Field welding or bolting joists to supporting steel framework shall be in accordance with the SJI specifications for the type of joist used, the erection/shop drawings and other Contract Documents.

The contractor shall make no modifications or repairs to steel joists without prior written approval.

Use of power-driven or power actuated fasteners in the diagonal and bottom chord members of the joists is not permitted.

5.3.9 *Floor Plate, Grating and Handrails*

Floor grating or floor plate and handrails shall be installed in the structure as soon as possible after the steel is erected to provide as much safe access through the structure.

The floor panels shall be fixed to the support steel immediately after placement using a minimum of four grating clips to secure each panel.

The clips shall be installed at 500 mm maximum intervals in direction at right angles to load bar along edge and intermediate supports, with a minimum of two clips per support.

All clip bolts shall be installed with a galvanized spring washer above the top plate.

5.3.10 *Coating Repairs*

The contractor shall repair all erection damage to hot-dip galvanized coatings. All erection damage to shop-applied paint coating shall be repaired in accordance with the coating specification. All galvanized bolt assemblies tightened against shop-applied paint shall be painted.

6. FIELD MODIFICATIONS

6.1 General

All field fabrication shall comply with the requirements of Steelwork Fabrication Specification, unless specifically modified below.

No modification likely to affect the strength of a structural member, the efficiency of a joint, or the integrity of a structure shall be made without the prior approval of the Employer.

6.2 Fabrication

6.2.1 *Cutting*

Wherever possible, site cutting shall be by a mechanical process. Where flame cutting is necessary, it shall be approved by the Employer and for such cases, an allowance shall be left for grinding off ragged edges to ensure a smooth finish.

6.2.2 *Holing*

Where additional bolt holes are required to be installed on site, the holes shall be drilled or punched. Flame cutting of holes for bolts shall not be permitted. Subject to prior approval by the Employer, correction of misaligned holes may be achieved by reaming or by filling badly aligned holes with weld metal, grinding flush and re-drilling in the correct position.

PROPRIETARY & CONFIDENTIAL

6.2.3 *Repositioning*

Where welded fitments need repositioning, the existing weld shall be removed where possible by chipping, grinding or gouging. When the fitment has been removed, any undercut areas shall be built up with weld metal and the affected area shall be ground smooth. Fitments damaged during removal shall be discarded and replaced as necessary using the same attachment method and detail as specified on the approved fabrication drawings or as directed by the Employer.

6.2.4 *Finishing*

Wherever remedial work has been carried out, the affected area shall be cleaned, burrs, slag and weld spatter removed and the surface treatment made good in accordance with the applicable Project Specification.

6.2.5 *Temporary Bracing Supports*

The Contractor shall remove temporary erection structures used during assembly and erection and grind surfaces smooth. Pits and gouges shall be fully welded to original metal thickness and ground smooth. All foreign substances shall be removed completely from all surfaces after completion of erection.

6.2.6 *Removal of Gussets*

Redundant existing gusset plates, stubs, etc. shall be cut off adjacent to their connection to the main member. Where they are attached to the web of a flanged member, only the portions outstanding from the sides of flanges shall be cut out, unless they affect the modification.

Redundant existing stiffeners not extending beyond the member need not be taken out unless the stiffeners affect the modification.

Where complete removal of the existing gusset plates, stubs, etc. is required, removal shall be carried out by cutting 3 mm away from the main member surface and grinding flush with the surface.

6.2.7 *Plugging of Bolt Holes*

Where redundant existing bolt holes impair structural performance, they shall be plugged with weld and ground smooth flush with the surface.

7. INSPECTION AND TESTING

7.1 General

A Quality Assurance/Inspection Plan shall be provided, detailing procedures to be employed and the responsible personnel, to ensure that the requirements of this Specification and the applicable codes and standards are met, as specified in MD-101-SMPM-PM-QA-PEI-0001, Contractor Quality Requirements.

Unless approved in writing beforehand by the Employer, all non-destructive testing (NDT) of welding shall be carried out by an independent registered Subcontractor. All NDT operators shall be experienced and shall be trained and certified to AWS QC1.

7.2 Inspection Requirements

The following inspections shall be carried out by the Contractor as a minimum:

7.2.1 *Materials*

Inspection procedures shall be developed to ensure that materials meet the requirements of Steelwork Fabrication Specification, prior to installation.

7.2.2 *Survey*

Structures shall be checked for compliance with Section 5.2.3.

A survey report shall be produced for all enclosures, towers and equipment supports indicating specified level and position, actual level and position, amount of non-alignment, and out of plumb and specified tolerances.

7.2.3 *Surface Treatment*

Visual inspection of surface treatment shall be carried out. Any instances of surface treatment impairment or damage shall be recorded, and all defects reported to the Employer before carrying out repairs.

7.2.4 *Field Welding*

Welding inspection and testing frequency for field welding shall be the same as specified for shop welding in Steel Fabrication Specification.

The Contractor shall specify in the ITP the methodology to be employed in the sampling and selection of the welder and the welds to be inspected.

PROPRIETARY & CONFIDENTIAL

7.2.5 *Bolting*

4.2.5.1. Snug Tight Type Bolting

The inspections shall confirm that:

- Members are drawn into firm and uniform contact, and
- A minimum of 10% of all bolts have been checked for tightness.

4.2.5.2. Pre-Tensioned Type and Slip-Critical Type Bolting

The inspections shall confirm that:

- Members are drawn into firm and uniform contact and the bolts are tightened to the tensions specified.
- A minimum of 10% of all bolts have been checked for tension using a calibrated torque wrench.
- An inspection (torque) wrench and appropriate calibration equipment is maintained by the contractor and tested each working shift. The contractor shall maintain test logs for the Employer's review.
- Any specific "hold" points specified in the contract are signed off by the Employer.